



Neutralizing the impact of atmospheric turbulence on complex scene imaging via deep learning

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A turbulent medium with eddies of different scales gives rise to fluctuations in the index of refraction during the process of wave propagation, which interferes with the original spatial relationship, phase relationship and optical path. The outputs of two-dimensional imaging systems suffer from anamorphosis brought about by this effect. Randomness, along with multiple types of degradation, make it a challenging task to analyse the reciprocal physical process. Here, we present a generative adversarial network (TSR-WGAN), which integrates temporal and spatial information embedded in the three-dimensional input to learn the representation of the residual between the observed and latent ideal data. Vision-friendly and credible sequences are produced without extra assumptions on the scale and strength of turbulence. The capability of TSR-WGAN is demonstrated through tests on our dataset, which contains 27,458 sequences with 411,870 frames of algorithm simulated data, physical simulated data and real data. TSR-WGAN exhibits promising visual quality and a deep understanding of the disparity between random perturbations and object movements. These preliminary results also shed light on the potential of deep learning to parse stochastic physical processes from particular perspectives and to solve complicated image reconstruction problems given limited data.

Turbulence arises as a result of the combined action of inertial force and viscous force on a flow field, where the damping effect of the viscosity is weaker than the kinetic energy of the fluid flow. The existence of eddies of different scales, and their interactions, make analysis of this a puzzle in classical physics¹. Owing to the complexity of the atmospheric boundary layer, turbulence is ubiquitous throughout this part of space, which has an intricate inner structure and complex transport properties². Thermally driven turbulence is widespread in routine life and in industrial production, as well as in surveillance and photogrammetry. The imaging process for the latter is presented in Fig. 1. The anisotropy and inhomogeneity in the possible multiple states of the airflow exacerbate fluctuations in the refractive index before light enters imaging equipment^{3,4}, and the different levels of geometric distortion and blurring affect the entire plane surface with spatiotemporally varying properties during short-exposure imaging⁵. In combination with atmospheric scintillation, these degradations are detrimental to situational awareness and subsequent data processing and analysis⁶. Numerous important applications, including long-distance security monitoring, laser communications and high-resolution Earth observation in the aerospace domain are seriously plagued by turbulent conditions. A robust route to turbulence mitigation is thus essential to meet the fundamental requirements of both military and civilian applications, and will also open up possibilities for exploring the characteristics of turbulent fields.

Numerous attempts have been made to mitigate the adverse effects of atmospheric turbulence, from hardware implementations to software processing. Adaptive optics systems, whose core components include a wavefront sensor and a deformable mirror, aim to correct the distortions introduced by atmospheric turbulence via high-frequency compensation of aberrations⁷. Together with speckle imaging and lucky imaging, these techniques go a long way

to achieving imaging with higher angular resolution for astrophotography^{8,9}. However, such adaptability in relation to astronomy image data from ground-based observations does not guarantee success in horizontal imaging of a complex scene, given that techniques such as adaptive optics are already usually implemented on large telescopes. Mainstream solutions for horizontal imaging through turbulence generally fall into two areas: single image reconstruction using multi-frame images to produce a static scene and dynamic scene restoration. Registration–fusion frameworks and blind deconvolution are the main schemes used for single image reconstruction, with massive efforts devoted to finding a solution for anisoplanatic errors and spatially variant blurrings^{5,10–12}. Although better results have been achieved compared to other methods, there is no denying the fact that using continuous frames of still images is impractical for most visually based applications. Attempts to employ extremely short exposures to freeze movements in sampling signals might also freeze the degradation from turbulence, and distortions cannot be corrected without a sufficient temporal reference. Hence, dynamic scene restoration has emerged from practical necessity. Techniques such as low-rank matrix decomposition and optical flow facilitate separation of a moving target and stationary background^{13,14}. However, degradation in targets and shifting imaging perspectives remain a problem.

Deep learning has been developing at a staggering rate in recent years. It has been used to tackle many unsolved problems in multiple fields, from bioscience to high-energy physics^{15–19}. Its powerful nonlinear mapping ability and massive parameters promise stable performance in processing high-dimensional data and modelling the inverse of certain intractable cases²⁰. Numerous reconstruction-related computer vision tasks, such as super-resolution and inpainting, have already been established by convolutional-network-based frameworks^{21,22}. However, as yet, there

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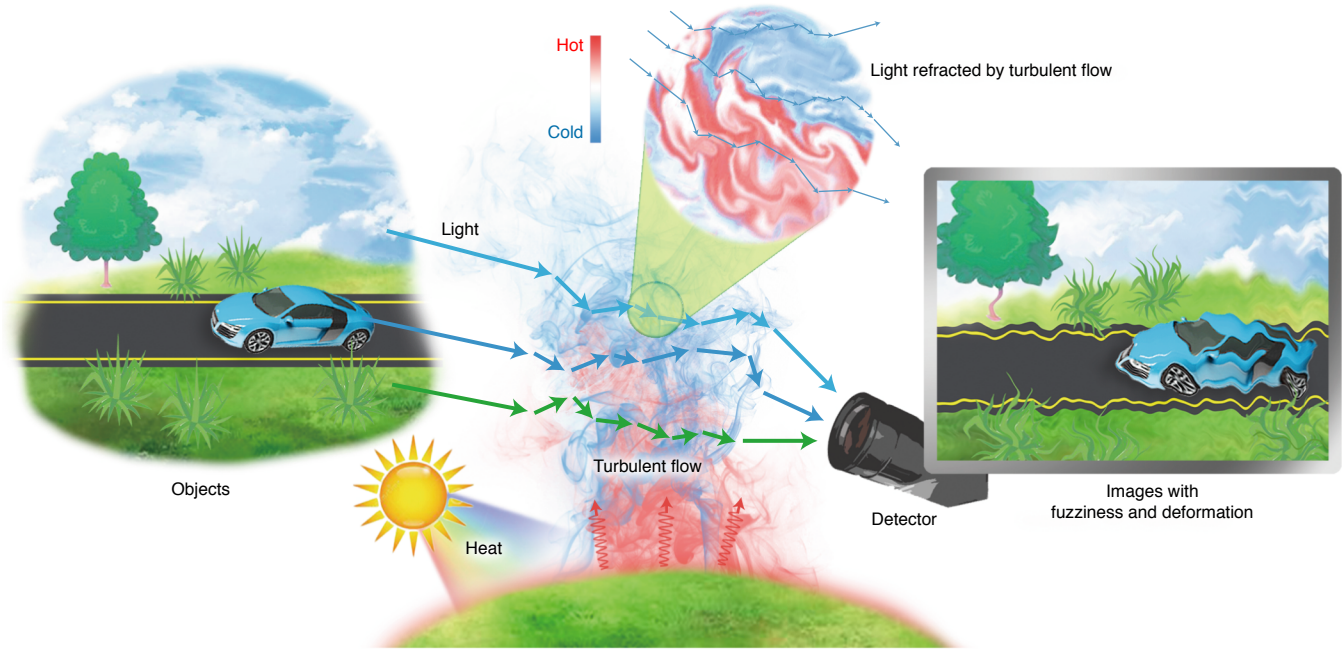


Fig. 1 | Schematic of imaging through atmospheric turbulence. An inhomogeneous air medium resulting from a high surface temperature in combination with complicated wind directions contributes to a disordered flow field. As a consequence of variations in the refractive index in different regions along the light path, the air can no longer be seen as a single medium, so refraction occurs during light transmission. There will therefore be an angle between the real light path and the ideal one, as a result of which the objects in the image plane are blurred and distorted in a spatiotemporally varying way.

have been no studies on parsing aerodynamic-phenomena-caused imaging problems using deep learning approaches. In turbulence-related degradation problems, according to the atmospheric refractivity spectral model, it is known that the spectrum of the turbulence, which behaves according to Kolmogorov statistics, takes the form $\Phi_n(k) = 0.033C_n^2k^{-11/3}$, where the refractivity structure index C_n^2 reflects the strength of the turbulence. For imaging devices, the modulation transfer function (MTF) can be formulated as $MTF = \exp(-\frac{1}{2}D(\rho))$, where D is the wave structure function deduced from the path integral of the refractivity spectrum²³. The above derivation reflects the contrast attenuation and strong non-linearity of the imaging process. We can thus construct the basic model for turbulence-related degradation in imaging as

$$o = B(MTF, x, y, t) \otimes (G(\xi(p, T, \lambda), \theta, \dots, x, y, t) \otimes z) + \epsilon,$$

where o , z and ϵ respectively denote the output image, latent clean image and random noise. $B(\cdot)$ and $G(\cdot)$ are the blurring and geometric distortion functions, $\otimes$ denotes the convolution, and (x, y) and time t demonstrate that spatially variant and temporally variant components exist in the degradation, as observed. The refractive index of air, ξ , the angle of arrival, θ , and a series of complicated factors such as the flow or movement of the air medium, changes in the light path and the imaging elements participate in the formation of distortion, as shown in Fig. 1. The mechanisms and forms of degradation in imaging caused by atmospheric turbulence differ from those of other kinds of image degradation, such as motion blur or out-of-focus blur. The multiple states of spatiotemporally varying geometrical distortion and blur during the turbulence-distorted imaging process are induced by disorder of the light propagation medium, while motion blur is caused by the relative movement of the targets and imaging devices. The noticeable distortion caused by turbulence is a geometric deformation that varies with both temporal and spatial positions. In combination with the blurring effect induced by chromatic dispersion and being

out of focus, turbulence-related distortion is far more complicated than the spatially invariant motion blur. As demonstrated above, turbulence-related imaging degradation processes are complex and nonlinear. The restoration of z may involve solving an underdetermined system of equations, which implies that multiple degradation processes would correspond to the same result. Constraints in the continuity of adjacent time intervals are necessary for solutions, contributing to the complexity. Neural networks are an effective tool for solving complex nonlinear problems, with the potential to fit the latent structure of the turbulence evolution process with a large number of parameters and a nonlinear activation function. Intuitively, targeted design with a network structure could lead to solutions to the anisotropy and random perturbations in imaging resulting from atmospheric turbulence.

To fill the gaps in compensating for atmospheric turbulence disturbance in imaging of a scene containing complex motion content, we propose our deep-learning-based methodology, termed TSR-WGAN (temporal-spatial residual perceiving Wasserstein generative adversarial network for turbulence-distorted sequence restoration). TSR-WGAN consists of three main parts that fit the turbulence imaging characteristics, the align module, the T-S integration module and a multiple discriminator design. Our method takes length-specified sequences as given data, aiming to construct a residual mapping from frames with temporal correlation to a designated label under adversarial mode. The T-S integration module is designed to integrate the spatial information of different time intervals and explore the underlying turbulence strength in a short exposure sequence by analysing the displacement of the pixels. The temporal attention-based align module and spatiotemporal conditional discrimination are implemented to reinforce continuity in the time domain, so as to preserve motions that are not relevant to atmospheric turbulence, which also extract multiscale features to fit the mechanism of turbulent movement. For training and testing, we built a turbulence-distorted video dataset comprising 27,458 videos with 411,870 frames, which includes algorithm

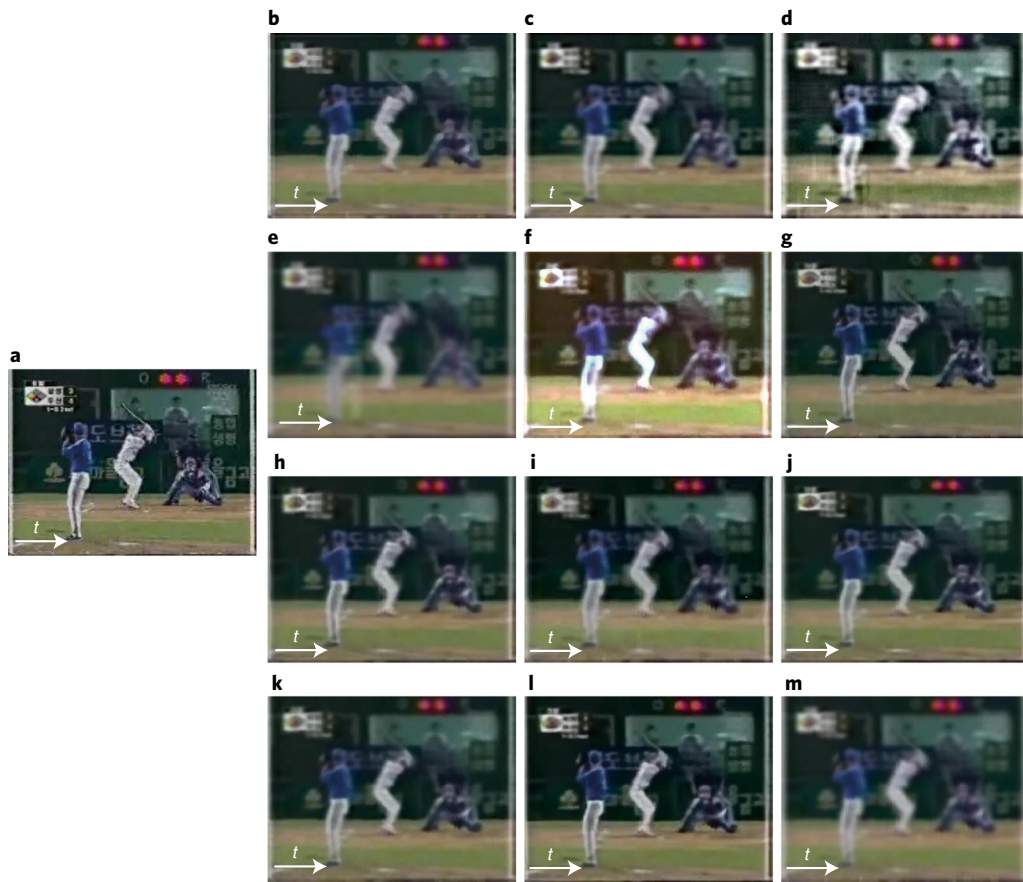


Fig. 2 | A comparison of turbulence mitigation models for both stabilizing and preserving motion information. A sequence of five columns is used to compose the shown results, with each image composed of 64 frames. **a**, The reference. **b–l**, The results for AL (**b**), SGF (**c**), CLEAR (**d**), LRP (**e**), DBN (**f**), DeblurGAN-v2 (**g**), STRCNN (**h**), DBLRGAN (**i**), CDVD-TSP (**j**), DVD-SFE (**k**) and TSR-WGAN (**l**). **m**, The composition of the original turbulence-distorted video sequence.

and physical simulated data and real data photographed under a climate with intense heat (in Australia). For comparison, a set of traditional turbulence mitigation approaches were also applied to demonstrate the generalizability of our method and its capability for nonlinear distortion correction, as well as the potential of deep learning for capturing the patterns of complicated physical phenomena. The prediction and control of turbulence is of great importance for aeronautics and astronautics, communication engineering and geoscience, for example, and applications such as laser communications, high-resolution Earth observation and climatic variation forecasting are greatly affected by turbulence. From the favourable quantitative performance and visual quality provided by the proposed turbulence mitigation technique, it is believed that related techniques have the potential to be effectively applied to analysing the properties and development of turbulent flow and solving key problems in the fields mentioned above.

Results

Atmospheric turbulence-distorted image dataset. We built a dataset containing algorithm and physical simulated turbulence-distorted video sequences and corresponding reference sequences, along with real-world data, to carry out an in-depth exploration of the patterns of behaviour of turbulence and facilitate assessment of the effect of characterizing the latent structure of turbulence. Specifically, our dataset contains 2,133 algorithm simulated sequences with 31,995 frames, 20,124 physical simulated sequences with 301,860 frames and 5,201 real-world sequences with 78,015 frames. The scenes in the dataset mainly consist of natural scenes such as the prairie and

desert, and urban scenes such as roadways and man-made structures, which contain large amounts of dynamic content. Details about the acquisition of data and the parameter settings are provided in the Methods.

Evaluation index. To effectively verify the performances of the various turbulence mitigation methods, a series of credible image quality assessment (IQA) and video quality assessment (VQA) measures were applied. IQA involves full-reference and reduced-reference measures to evaluate data with labels appropriately. The full-reference measures include the peak signal-to-noise ratio, the structural similarity index (SSIM)²⁴, gradient magnitude similarity deviation (GMSD)²⁵, feature similarity (FSIM)²⁶ and the visual saliency-induced index (VSI)²⁷, which are used to provide an assessment of a single frame in relation to perceived visual quality. Temporal continuity is also important for results evaluation in our experiments. The VQA reduced reference entropic differencing (RRED) index²⁸, including S-RRED and T-RRED, was used to perform integration of three-dimensional assessment scores.

Given the limitations of current objective evaluation systems regarding degradation, mainly in relation to compression and transmission, a subjective survey is necessary for validation of the effectiveness of turbulence mitigation. The results obtained from turbulence-removal methods, along with the original test data, were randomized and sampled in pairs for comparison. Human participants were asked to choose a video sequence, from each pair, that they thought to be of high quality, without the effect of turbulence. The survey included 10,000 pairs of video sequences and

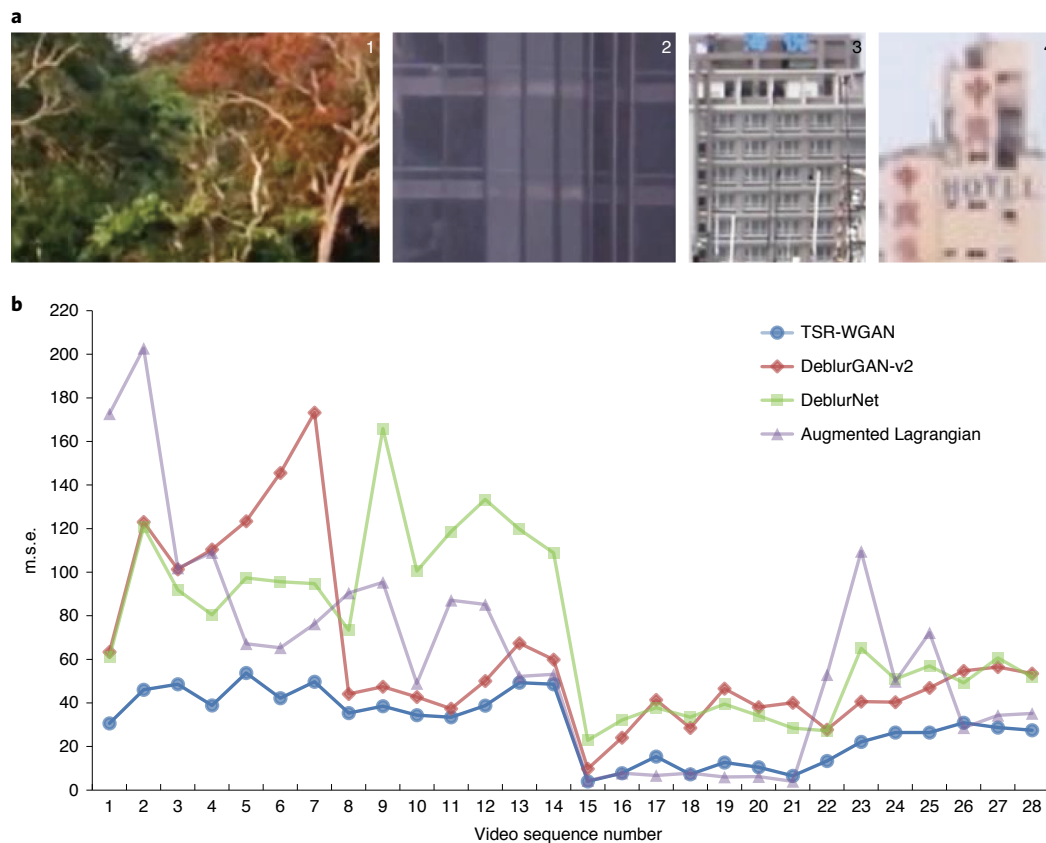


Fig. 3 | A comparison of sequence-processing methods on a static region. a, Static images sampled from the physical simulated dataset. The resolution of the scenes is 161×123 px (image 1), 137×220 px (image 2), 189×233 px (image 3) and 77×60 px (image 4). **b**, Mean square error (m.s.e.) scores of static regions of 28 physical simulated video sequences processed by TSR-WGAN, DeblurGAN-v2, DeblurNet and AL methods, where 1–7, 8–15, 16–21 and 22–28 represent the same scenes but with different turbulence strengths, corresponding to images 1, 2, 3 and 4 in **a**, respectively.

there were 50 human participants, each of whom compared 200 pairs. Most of the participants were from colleges and universities (for example, Beihang University, Beijing University of Posts and Telecommunications and Southwestern University of Finance and Economics). The comparison results were fit using the Bradley–Terry model²⁹ to obtain a true rating for each method. Software will be released for subjective pair comparison; this will integrate the functionalities of data distribution, comparison and collection (details are provided in Supplementary Fig. 1).

Comparison methods. Traditional registration–fusion framework, frequency-domain filtering and improved lucky-region methods were used for comparison (specifically CLEAR¹⁰, the Laplacian–Riesz pyramid method (LRP)³⁰, the Sobolev gradient flow method (SGF)³¹ and the augmented Lagrangian (AL) methods³²). As well as unsupervised methods, deep-learning based methods for deblurring, including DeblurNet³³, DeblurGAN-v2³⁴, DBLRGAN³⁵, STRCNN³⁶, CDVD-TSP³⁷ and DVD-SFE³⁸, were also compared to demonstrate the power of the proposed TSR-WGAN. Full temporal sequences were adopted as input for the LRP, SGF and AL methods according to the requirements of the corresponding algorithms. For the single-frame reconstruction method CLEAR, the input was divided into sequences of 15 continuous frames, with adjacent sequences having 14 frames overlapping one another. The corresponding parameters followed the default settings of the open-source codes. For deep-learning based methods, the default settings of their hyperparameters were adopted, as listed in Supplementary Table 1.

Performance on algorithm simulated data. Experiments on algorithm simulated data were conducted to evaluate the performance of the turbulence mitigation methods on both pure static scenes and dynamic scenes with a greater diversity of disturbance settings. Details about the algorithm simulated data are provided in the Methods. The ratio of static and dynamic scenes was 15:9, with each scene processed by the algorithm with three different parameter settings to enrich the variations in the strength of the turbulence. Quantitative results are presented in Supplementary Table 2. With the exception of RRED and GMSD, higher values mean better visual quality. The values of the evaluation indices in the table are mean values of the algorithm simulated test data. TSR-WGAN shows a clear advantage for all six evaluation indices. Generally speaking, models aiming at single-frame restoration tend to give favourable results for static scenes. However, our experiments indicate that TSR-WGAN is more suited to stabilizing and handling blurs of different degrees caused by turbulence. Visual results are presented in Supplementary Fig. 2. It is observed from the zoom-in view in this figure that the proposed TSR-WGAN and SGF cope better with blurs when keeping the original hue. In the results for CLEAR and DBN, the details are distinctly enhanced, but the sharpening process is excessive and there is a large deviation in the colour space. The other methods fail in detail reconstruction.

TSR-WGAN exhibits a clearer advantage in the reconstruction of dynamic scenes (Fig. 2). Five continuous columns were sampled from the corresponding positions in a frame (Fig. 2a), and the results were composed from a total of 64 frames (Fig. 2b–l). Blurring is still severe in the results achieved by the AL, SGF,

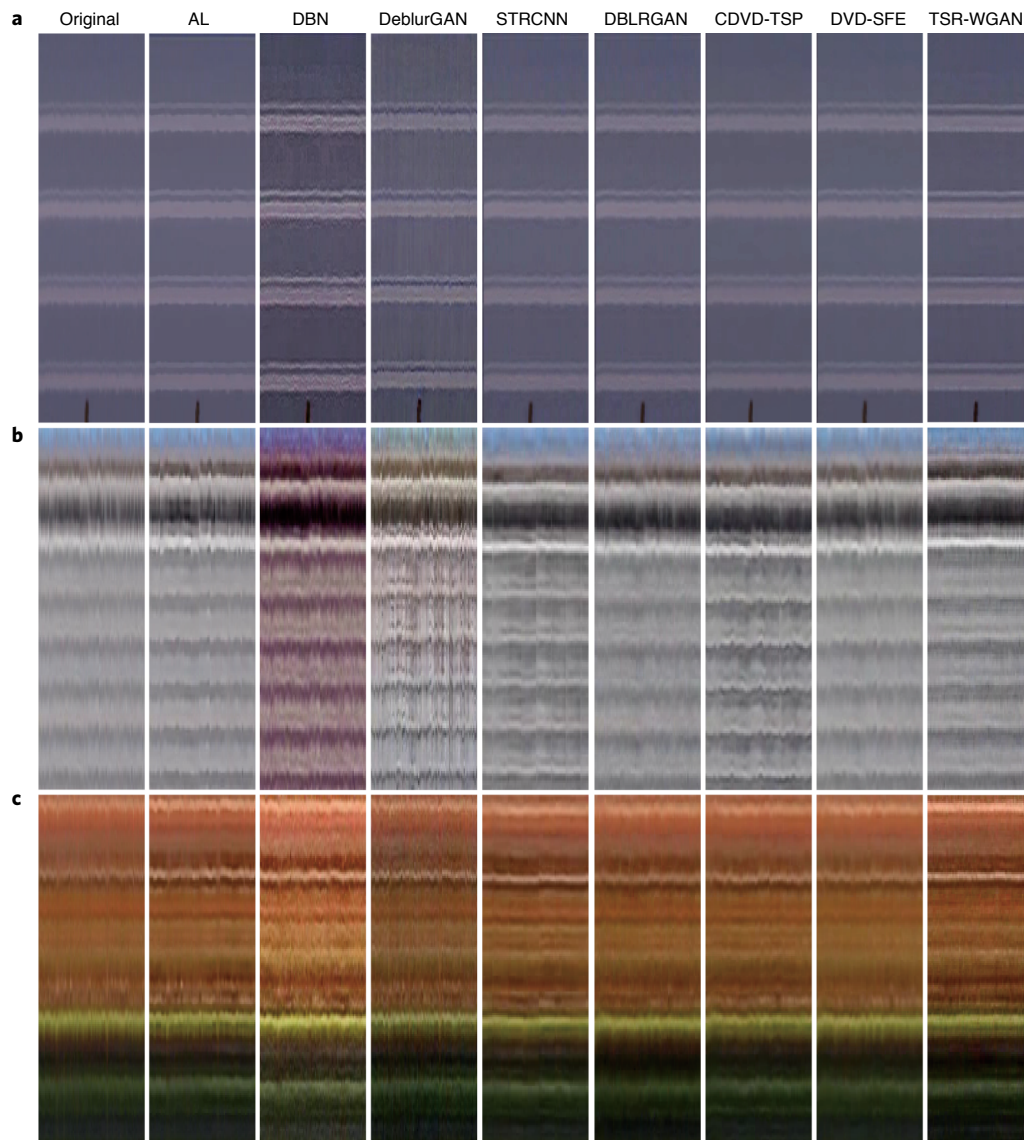


Fig. 4 | Column sequences with the same temporal index. Selected columns of static regions from images in Fig. 3, arranged in temporal order to visually demonstrate stability. **a**, The 122nd column sequences of the second scene in Fig. 3a, produced by the methods indicated at the top. **b**, The 77th column sequences of the third scene in Fig. 3a. **c**, The 159th column sequences of the first scene in Fig. 3a.

CLEAR and LRP methods (Fig. 2b–e), and there is even smearing in the sampling result with the LRP method. For most of the other deep-learning-based deblurring frameworks, the clarity is limited and there is distinct distortion in the rectangular score board in the top left of the image, which indicates that geometric distortion exists in this region and changes over time. Furthermore, the moving parts of the video sequence—mainly people—have serrated edges because of the vibration in particular regions. The proposed TSR-WGAN method better distinguishes between the distortion caused by turbulence and the motion in the content so that it simultaneously retains fluent and clear motion and removes the turbulence effect. More dynamic results are provided in Supplementary Videos 1–3. It is evident that flutter in the content is eliminated and details are improved by TSR-WGAN compared with the other methods, with temporal coherence simultaneously preserved.

Performance on physical simulated data. All comparison methods were run on physical simulated data. Details about data acquisition are provided in the Methods. Supervised methods, including

TSR-WGAN, DeblurGAN-v2, DeBlurNet, STRCNN, DBLRGAN, CDVD-TSP and DVD-SFE, were trained on a training set sampled from our turbulence-distorted dataset (other than the test data). The volume of the training set was 10,457, roughly equal to that of the test set (9,667). This was intentional to prevent overfitting phenomena caused by disparity between the sizes of the training and test data. Based on the ground truth of physical simulated data, full-reference evaluation indices were applied. The quantitative results are reported in Supplementary Table 3. The values of the indices in the table are mean values of all the physical simulated test data. The proposed TSR-WGAN achieved the best quantitative performance on common objective image/video quality assessment. Notably, the statistical performance of TSR-WGAN was significantly better than that of the compared methods, with a value 45.69% lower than the average comparison level for the considered temporal coherence indices. The temporal RRED index is designed to measure motion differences between a distorted video and a reference, and is sensitive to the typical scintillation and geometric distortion in turbulent scenes. This shows that TSR-WGAN

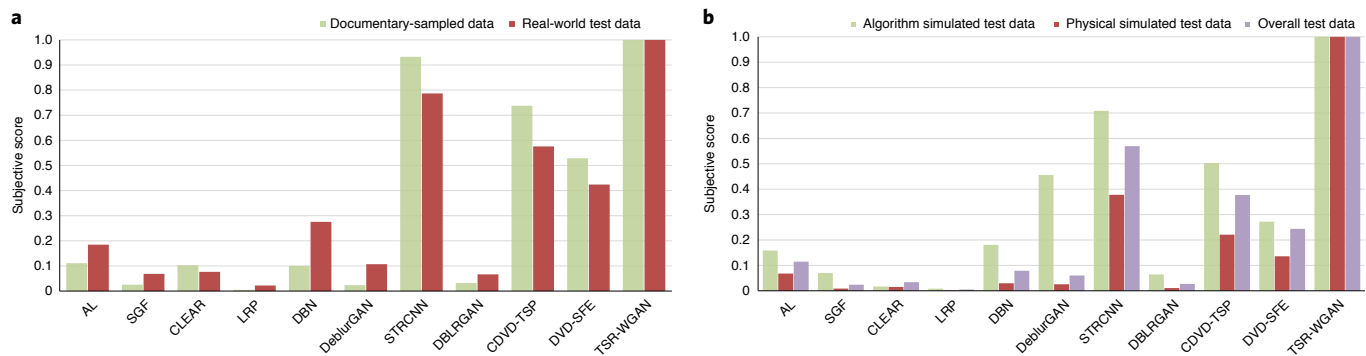


Fig. 5 | A subjective comparison of the turbulence-mitigation models. a, Subjective scores of TSR-WGAN and the other methods on real-world test data and documentary-sampled data. **b**, The overall scores of each method on the entire test set, including physical simulated, algorithm simulated and real-world data.

performs effective stabilization of fluctuations in content and mitigates visual discontinuity. Supplementary Fig. 3 visually shows the turbulence-removal capability of TSR-WGAN. Consistent with the quantitative results, the proposed TSR-WGAN visibly suppresses the blurring effect brought about by turbulence.

Because of the multiple changes in the angle of arrival arising from the non-homogeneous medium, anomalous glint will exist in both static and moving objects. Thus, unlike in the deblurring task, stabilization is also an important factor in turbulence neutralization. Static regions are sampled from video sequences to compute the variance of pixel intensity over the entire duration time, as presented in Fig. 3. More stable performance is offered by TSR-WGAN, with the mean square errors of the results being significantly lower than for the other methods on the tested scenes. Figure 4 shows the differences in stabilizing the static content of the various methods. It can be seen that the sawtooth-like wave caused by atmospheric disturbance is quite mitigated by TSR-WGAN, whereas improvements by the other sequence-processing methods are not apparent—with the other methods, the lines remain unsmooth, indicating there is jitter, in agreement with the numerical results. The CLEAR, LRP and SGF methods are not compared in this task because these single-frame-targeted frameworks have difficulty in restoring fluent motion. The results are usually accompanied by serious smearing problems, as shown in Supplementary Fig. 4. Further dynamic results are available in Supplementary Videos 4–6. In dynamic scenes, single-frame-based methods fail to restore object motion, whereas TSR-WGAN better distinguishes movement of the content from fluctuations and scintillation caused by turbulence, thus giving satisfactory visual results.

Performance on real data. The real data are composed of real-world video sequences recorded in Australia and scenes sampled from a series of documentaries (details are presented in the Methods). Because of the absence of ground truth, we adopted a subjective rating strategy. Figure 5a demonstrates the generalization performance of TSR-WGAN, which shows true rating scores that are substantially higher than those of the other methods. It can thus be inferred that the TSR-WGAN model, trained on the physical simulated data, has captured a universal pattern of the turbulence effect in the imaging results, and could effectively neutralize or compensate for the related distortions based on spatial and temporal relativity. More results are provided in Supplementary Fig. 5, and dynamic demonstrations are presented in Supplementary Videos 7–9. The changing depth of the turbulence field complicates the distortion in the imaging plane, where the spatiotemporally varying properties are intensified. TSR-WGAN still effectively suppresses the turbulence effect

and from this we can infer that the capacity of the proposed method could be further improved with broader data sampling.

We also verified the performance of the tested methods on algorithm and physical simulated test data by subjective investigation using the Bradley–Terry model. The subjective scores of TSR-WGAN are still significantly higher than the other methods (Fig. 5b). From analysis of the subjective and objective evaluation results, human observers can be seen to show a greater preference for the results processed by TSR-WGAN than the numerical results given by traditional image/video quality assessment. This is because human visual systems focus more on interframe continuity and are more sensitive to the local irregular variation introduced by turbulence. Several commonly used performance metrics for evaluating IQA were adopted here to explore the correlation between these indices and human perception. We utilized the Pearson linear correlation coefficient (PCC), the Spearman rank order correlation coefficient (SRC) and root mean square error (r.m.s.e.) to conduct a correlation analysis between the indices and human subjective scores. The values of the PCC, SRC and r.m.s.e. range between 0 and 1, with higher values indicating that the IQA indices have a better consistency with human visual systems. The results are demonstrated in Supplementary Fig. 6. The analysis results indicate that the degree of matching of classical full-reference IQA measures and human visual systems on the form of atmospheric turbulence distortion is limited. It also reveals a latent demand for an evaluation index for this kind of degradation.

In conclusion, the proposed TSR-WGAN approach has been compared with existing classical turbulence-mitigation methods and supervised deblurring methods to verify its efficacy. We built an atmospheric-turbulence-distorted video sequence dataset comprising a total of 27,458 videos with 411,870 frames for modelling and experiments, which contains paired algorithm and physical simulated data (Fig. 6) and real-world turbulence-containing data. Classical full-reference/reduced-reference image/video quality assessment measures were adopted to evaluate the performance of the tested methods on turbulence mitigation. Human subjective investigation via pair comparison was conducted to analyse the preference of human visual systems in relation to the turbulence removal effect. The numerical results for both objective and subjective indices demonstrate the superiority of TSR-WGAN over the other methods. Furthermore, the TSR-WGAN model used in the tests was trained only on physical simulated data and achieves promising results on both algorithm simulated and real-world data, supporting the generalizability of the method. The prominent performance of TSR-WGAN suggests promising applications in surveillance, remote sensing and photography with mobile devices in

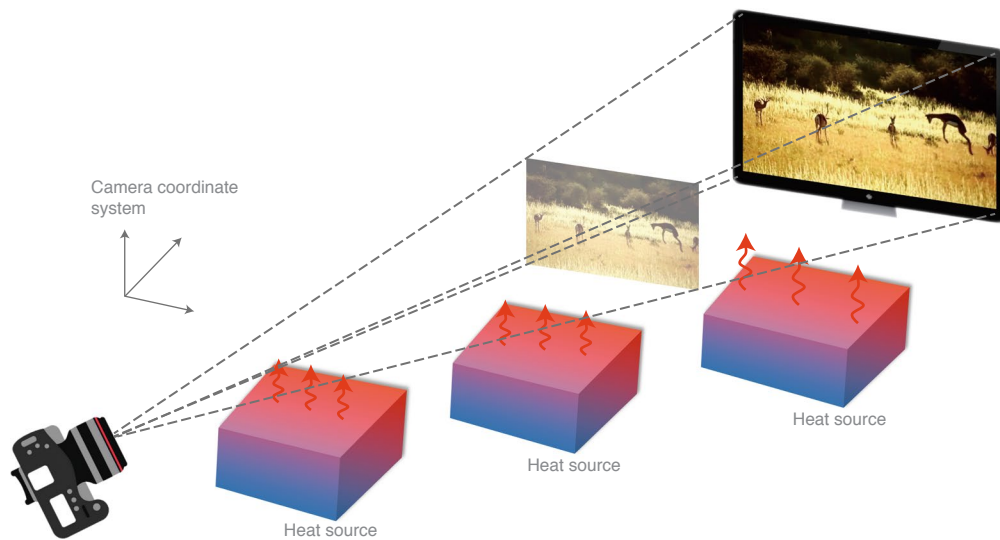


Fig. 6 | A sketch of the physical simulation platform. The physical simulation platform, in brief, comprises a camera, heat sources and a display device.

daily use, which could effectively improve the visual quality of data recorded at comparatively long distances. Our research shows that deep-learning-based methods with well-designed architectures, which are parameter-independent, have the potential to cope with inverse imaging problems for complex stochastic aerodynamic phenomena, without complicated handcrafted features. It is conceivable to further develop related techniques and theoretical models to tackle key application problems in several military and civilian areas, including coherence degradation in laser communications, visibility enhancement in high-resolution Earth observation, as well as further exploration of turbulence mechanisms in chaos theory studies and modern industrial applications.

Methods

Acquisition of algorithm simulated data. We adopted a heuristics-based turbulence-simulating method to obtain the algorithm simulated data, and improved generalization to diverse scenarios and turbulence of different degrees³⁰. The simulation works in a two-dimensional way. Given appropriate property settings for the simulated turbulence model and imaging systems, including an atmospheric refractive index structure parameter, inner scale, outer scale, aperture diameter, focal length and imaging distances, a point spread function producing spatially variant blur could be generated. Combined with the data provided in ref.³⁹ recorded at White Sands Missile Range, the angle-of-arrival variances of spherical and plane waves were computed to carry out random excursion for the imaging points. Given the potential interframe discontinuity introduced by stochastic numerical simulations, an interpolation scheme was utilized to compensate the distortion along the time index. By synthesizing spatiotemporally variant blur and typical deformation patterns, we generated an algorithm simulated dataset comprising 2,133 video footages with 31,995 frames, which originated from the Berkeley Segmentation Data Set and Benchmarks 500 (BSDS500)⁴⁰ and UCF101⁴¹. In a specific simulation, the atmospheric refractive index structure parameter C_n^2 was randomly sampled from 10^{-16} to 10^{-12} to guarantee data variability, along with imaging distances d ranging from 10 m to 2,000 m and multiple focal lengths f , with the different parameter settings and contexts allowing us to verify the generalization of the test models.

Acquisition of physical simulated data. A range of real-world scenes involving turbulence, with corresponding views through a homogeneous medium, were required to facilitate parsing of the imaging process, bridging patterns with complicated physical mechanisms and fully validating the efficacy of the different models. Given that strictly matched dynamic scenes are difficult to obtain in the real world, physical simulation under controlled conditions is necessary to train and evaluate restoration models. A physical simulation platform was thus built to record particular scenes disturbed by turbulence of various intensities, as well as a corresponding blank group that consisted of imaging equipment, the heating apparatus and devices for playing the content. Six gas burners with power of 3.5 kW were utilized to generate turbulence. Every two gas burners were juxtaposed with each other, and the three pairs were placed equidistantly with a spacing of 1.8 m. A Canon EOS 5D Mark IV camera was set 1.5 m away from the last pair of gas burners.

The camera was used to record content playing on a 27-inch display screen (placed 1.5 m in front of the first pair of gas burners) with a resolution of $1,920 \times 1,080$, through turbulence. Figure 6 shows the basic structure of the physical simulation platform. Fixed position and parameter settings were adopted for the SLR camera (focal length of 400 mm, shutter speed of 1/100, aperture size of 1/5.6 and ISO of 1,250). From a series of documentaries produced by the BBC, Disney and so on, we sampled 20,124 video sequences containing both natural scenes and urban scenes. Wild animals, various kinds of vegetation, human activities, man-made buildings and vehicles were included to enrich the variety and complexity of the content. The recording process was conducted in the laboratory with a stable artificial light source. We adopted eight settings to shoot the content on the display screen, with each pair of gas burners in a binary state, giving 2^3 combinations. The built physical simulated dataset thus contained 301,860 turbulence-distorted frames and a corresponding latent clean reference, with turbulence of varying intensity and different scenes effectively verifying the generalization of the model.

Acquisition of real data. To fully demonstrate the performances of the different models in neutralizing the impact of turbulence, we constructed a dataset composed of 5,201 real-world video sequences with 78,015 frames containing thermally driven turbulence. The real-world data were composed of two parts. The first contained 4,303 video sequences shot in Australia and 898 clips containing turbulence sampled from a series of documentaries. In Brisbane, Australia, we used a Nikon D750 camera to capture images in street view when the local temperature was $\sim 93^\circ\text{F}$ to 99°F . Among the collected data, 4,303 video sequences involving typical scenes (including buildings, vehicles and pedestrians) were sampled and constitute the second part of the dataset. These were mainly recorded in Queen Street ($153^\circ 2' \text{E}$, $-27^\circ 28' \text{N}$), Victoria Bridge and Pinelands Road ($152^\circ 45' \text{E}$, $-27^\circ 35' \text{N}$). The related camera parameter settings were from auto mode, with shutter speed of 1/250–1/200, aperture size of 1/11–1/9 and ISO of 100–300. The resolution of the original video was $1,920 \times 1,080$, and the sequences used in the dataset were cropped to an appropriate size. Real-world turbulence, which is influenced by various complicated factors as well as moving objects at different distances, could further validate the efficacy of the proposed and comparison models.

Network architecture. The architecture of our TSR-WGAN generator, presented in Supplementary Fig. 7, comprises two functional modules designed based on a turbulence imaging mechanism, namely an align module and a T-S integration module, as well as an encoder–decoder architecture. The scale of turbulence spans a wide range, which can be roughly divided into large-scale and small-scale turbulences. The large-scale turbulence dominates momentum and scalar transport, and its size is comparable to the flow field. The small-scale turbulence dissipates the energy transported from large-scale turbulence. When light propagates through a field, it is simultaneously influenced by turbulence of all scales. Owing to the variation in the scales and anisotropy, the modulation transfer function (MTF) and angle of arrival (AOA) are formulated as $\text{MTF}(\dots, L=l, i=1, 2, 3, \dots), x, y$ and $\text{AOA}(\dots, L=l, i=1, 2, 3, \dots), x, y$ where L is the scale-related variable. Focusing on the comprehensive effect of scales and spatial variation, we designed the align module with a cascaded connection to sample features from the input sequences in a multiscale manner, to benefit the perception of geometric distortion and blur introduced by turbulence of different scales. The corresponding function can be expressed as $\text{Out} = F(\text{input}, F(\text{input}_{L=\frac{1}{2}}, F(\text{input}_{L=\frac{1}{4}})))$, where

$F(\cdot)$ is the feature extraction, composed of convolution, ReLU (rectified linear unit) and identical mapping. The internal structure of the module mainly consists of a cascading block that extracts features of different scales and a temporal attention unit that reweights the upstream feature maps of the frames in accordance with their responses to the central frame and thus aligns the sequence. The turbulence-perturbed sequences are segregated and fed into the align module. The aligned 4D feature maps are divided into several groups (usually N groups if there are $2N-1$ input frames) to undergo a 3D convolution operation in the T-S integration module, which freezes the motion in a short time interval, like using short exposure to facilitate a stable temporal observation.

For image jittering related to changes in the angle of arrival, one can use the analytical model $\langle \beta^2 \rangle = 2.91 f^2 D^{-\frac{1}{2}} L C_n^2$ (ref. ⁴²), where $\langle \beta^2 \rangle$ is the mean square image displacement, C_n^2 can be inferred from the statistic of pixel displacement, f is the focal length, D the aperture size and L the optical path length. Specifically, the form should be $C_n^2 = 0.34 f^{-2} D^{\frac{1}{2}} L^{-1} \langle \beta^2 \rangle = A \langle \beta^2 \rangle$. Thus, the turbulence-strength-related variable C_n^2 can be expressed in terms of $\langle \beta^2 \rangle$, which can be computed as a time series. Based on the analysis above, we designed the T-S integration module to give an insight into correlation among the pixels in temporal and spatial order, which is expected to estimate the turbulence strength. Taking the motion of the imaging content into consideration, a temporal attention mechanism is applied before the T-S module to distinguish the motion of the target from turbulence perturbation, which should promote a more accurate displacement feature extraction. This kind of design aims to explore possible dynamic mapping among the different temporal intervals, rather than conducting a 3D convolution operation with the same depth on the ungrouped features. The constructed feature layers, following 3D convolution, are concatenated in chronological order and fused into dimension-reduced maps for further 2D tasks. As for the length of the temporal window in the T-S integration module, there is a rationale where each divided subinterval of temporal sequences should contain the centre target frame. This rationale is consistent with the target of reconstructing the centre target frame, where spatiotemporal feature extraction in each subinterval involves the target frame. Thus, the length is always set as $\frac{2N+1}{2}$, where N is the length of the input sequence. Each subinterval is obtained by sliding the temporal window one frame each time, and the number of the divided subintervals is correspondingly $\frac{2N+1}{2}$.

The encoder-decoder architecture, which provides a multiscale analysis to fit the nature of the turbulence scenario, is used to produce a degradation-neutralized result from the generated feature maps. The structure is inherited from SegNet⁴³ and DeblurGAN⁴⁴. A symmetrical structure involving strided and transposed convolution promises consistency of resolution, where residual blocks⁴⁵ balance the depth of the network and the training. Simultaneously, a global connection leading to a summation of the central input frame and the learned residual map markedly accelerates the convergence process and facilitates giving a stable generative model. The structures of the align module and T-S module are described in Supplementary Figs. 8 and 9.

A multiple discriminator design is used for TSR-WGAN to augment reconstruction of different scales. The architecture of the single discriminator is adapted from that of PatchGAN⁴⁶ and CycleGAN⁴⁷. We substitute the convolution operation with dilation convolution for a larger receptive field and delete some of the layers for the local discriminator, considering the size of the output. A temporal conditional discrimination strategy is implemented during the adversarial process, where unperturbed content from neighbouring time points clamps the generated frame or its corresponding ground truth, giving a sandwich-like shape for discrimination. A random search process is applied to obtain local patches with 1/8 resolution as sources for local discrimination to enhance detail reconstruction. At the terminal, the mean values of the concatenated outputs of the global and local views are regarded as a score for judging true or false. The architecture of TSR-WGAN's discriminator is shown in Supplementary Fig. 10.

Perceptual loss⁴⁸ and WGAN-GP⁴⁹ (GP, gradient penalty) loss are utilized as the critic function for our network:

$$L = \lambda L_{\text{feat}} + L_{\text{MSE}} + L_{\text{GAN-g}}$$

where L_{feat} is the perceptual loss weighted by parameter λ and $L_{\text{GAN-g}}$ is the WGAN-GP loss for the generator. The loss functions are defined as follows:

$$L_{\text{feat}} = \frac{1}{N_j} \|\Phi_j(G(z)) - \Phi_j(x_i)\|^2$$

$$L_{\text{GAN-g}} = -\mathbb{E}_{G(z) \in \mathbb{P}_g} [D(G(z)|x_{t-1}, x_{t+1})]$$

$$L_{\text{GAN-d}} = \mathbb{E}_{G(z) \in \mathbb{P}_g} [D(G(z)|x_{t-1}, x_{t+1})]$$

$$-\mathbb{E}_{x \in \mathbb{P}_r} [D(x|x_{t-1}, x_{t+1})] + \eta \mathbb{E}_{\hat{x} \in \mathbb{P}_{\hat{z}}} [(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2]$$

where $L_{\text{GAN-d}}$ represents the loss function for the discriminator. Here, x represents the real data and z follows the distribution influenced by turbulence. The estimated $\hat{x}$ from the generator is also expressed as $G(z)$. x_i is the corresponding clean reference of the i th frame of the input video, which is the centred target frame of input z . $\Phi(\cdot)$

denotes an output produced by the top j convolution layers of a pretrained network, which was chosen to be VGG19's top 14 layers pretrained on ImageNet Dataset.

Ablation study. Ablation experiments were conducted on the proposed align module, T-S module and the multiple discriminator design to demonstrate the corresponding benefits for turbulence neutralization. Specifically, five different settings were used to prove the efficacy of the proposed module. The numerical results are shown in Supplementary Table 4. We gradually added the proposed modules into the network architecture. It can be concluded from the tables that these modules steadily boost the performance on the peak signal-to-noise ratio for physical and algorithm simulated data. The strategy applied in Fig. 4 was adopted to demonstrate the stabilizing effect of the models for ablation. As shown in Supplementary Fig. 11, compared with the results of the net equipped with integrated modules, the lack of a module causes more distinct sawtooth-like waves in the local structures, indicating instability of the imaging content.

We also conducted a visualization of temporal attention. The attention channel of the align module is presented in Supplementary Fig. 12. It is noticeable that the highlighted part in the attention map in Supplementary Fig. 12c corresponds to the motion of the targets, which is key to turbulence neutralization in dynamic scenes. Decoupling the motion and turbulence-related perturbation in the imaging process leads to a better performance on time-varying geometric distortion suppression and continuity preservation. This indicates that the trained module captures the intrinsic distinction between the two kinds of movement and provides a strong prior for downstream inversion inference.

Implementation details. Our TSR-WGAN was implemented using PyTorch⁵⁰. The training process was conducted on a platform with an Intel i9-9900K central processing unit and GIGABYTE GeForce RTX 2080Ti with 11 GB of graphics memory and 32 GB of random access memory. The dataset used for training contained 10,457 video sequences randomly sampled from the physical simulated data. The corresponding frames were cropped to a size of $200 \times 200 \times N$. Before the start of training, a normal distribution with zero mean and 0.02 standard deviation was determined to initialize the weights of the convolution kernels. An Adam optimizer was utilized to update the weights with initial learning rates of 3×10^{-4} and 1×10^{-5} , respectively, for the generator and discriminator. The optimizer's β_1 and β_2 values were set to $\beta_1 = 0.9$ and $\beta_2 = 0.999$. As a result of the existence of perceptual loss, which is supervised, a smaller learning rate was applied for the discriminator. This is done because the domination of the supervised loss function drives more stable training and faster convergence, which is a deep-rooted problem for GAN. The training loss curves of the generator and discriminator are plotted in Supplementary Fig. 13. The steady training process further justifies our learning rate setting. The batch size was set to three and the training step ratio of the generator and discriminator was set to 1:1. It took 100 epochs for the proposed model to converge. Using our platform, it was determined that the typical inference time of TSR-WGAN at a resolution of 960×540 is 0.0777 s per frame, leading to high performance while maintaining high efficiency.

Broader impact statement. The analytical method discussed in this work is transferable to related domains where deep-learning models can be used for tackling complicated physical processes involving inverse problems. The strategy may be extended to practical applications such as parsing the physical properties of a turbulent flow field. The proposed TSR-WGAN framework can be applied to a wide range of uses. For instance, it can be applied to the receivers for laser communication or long-distance surveillance systems, which are severely influenced by the effects of atmospheric conditions on signals. The applications described above play an important role in industrial settings and have great societal importance.

Solving inverse problems can lead to a potential risk in terms of privacy. The inversion of reasoning in both the physical and digital world is fuelled by huge amounts of data, and there is a risk of breaches of physical shields or digital encryption by inverse inference based on specific remaining features. Because of these risks, we recommend that society and the scientific community focus on ethical issues related to data acquisition and usage, such as informed consent and permissible use.

Data availability

The full datasets, including the algorithm simulated data, physical simulated data and real-world data presented in this Article are publicly available in the Zenodo repository at <https://doi.org/10.5281/zenodo.5101910>⁵¹.

Code availability

The code presented in this Article is available through a Code Ocean compute capsule (10.24433/CO.3517894.v1)⁵², together with a subset of data to test the network.

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Author contributions

D.J. and X.B. conceived the idea and were responsible for the methodology. D.J., Y.C., Y.L., Z.L. and S.G. performed the experiments and created the dataset. D.J. and J.C. developed the software and designed the subjective evaluation. Y.C. and P.W. participated in the model design. X.B. supervised the project. D.J. and X.B. wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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